

MPFNet: Multiscale Perception Fusion Network for Multispectral Object Detection

Fanyou Zhou[✉], Jie Hu^{*✉}, Bo Peng[✉], *Member, IEEE*, Shengdong Du[✉], Tianrui Li[✉], *Senior Member, IEEE*

Abstract—Multispectral object detection has shown significant advantages in various environments by combining the complementary characteristics of the two modal images. However, existing methods still face challenges in multi-scale feature representation and the fusion of complementary multimodal information. To address these challenges, this paper proposes a multiscale perception fusion network named MPFNet for multispectral object detection. Specifically, at the feature extraction stage, we embed a multi-scale perception convolution module (MPC), which progressively organizes convolutional kernels of varying sizes to enhance multi-scale object perception. At the feature fusion level, the dynamic adaptive fusion module (DAF) is designed to achieve intelligent weighting between modalities through bidirectional cross-attention. Finally, to further generate higher-quality fused features, we add a cross-modal contrastive consistency (CMCC) loss to directly guide the fusion process. Comprehensive tests on the DroneVehicle, M3FD, and FLIR public datasets demonstrate that MPFNet works better than current techniques, providing accurate and reliable object detection in challenging situations.

Index Terms—Multispectral feature fusion, RGB-T object detection, multi-scale feature enhancement, attention mechanism.

I. INTRODUCTION

OBJECT detection, a core technology in computer vision, has a wide range of practical applications, including autonomous driving, video surveillance, remote sensing, and robotics. Object detection techniques based on visible light (RGB) images have advanced significantly with the quick development of convolutional neural networks and Transformer. However, the performance of RGB-based object detectors degrades rapidly under challenging conditions, such as rainy days, hazy weather, and low illumination [1], etc. Simply relying on a mono-modality for object detection is no longer sufficient to meet the practical application requirements. Compared with RGB images, thermal images rely on the emitted

radiation of objects, enabling it to produce high-quality images and perceptible contour features even in challenging environmental conditions. Therefore, multispectral object detection through the fusion of RGB and thermal images has received increasing attention [2], [3]. By combining the two modes and effectively leveraging their complementary advantages, the reliability of object detection in complex scenarios has been significantly enhanced [4], [5].

However, there are still two main challenges in the detection methods for RGB-thermal images. On the one hand, although multispectral images provide more complementary information for object detection, the different imaging principles of the two modalities result in differences in features and changes in target scales [6]. Therefore, the backbone network struggles to extract object features deeply, often leading to missed detections, as shown in Fig. 1 (a) and Fig. 1 (b). The current methods have obvious limitations when dealing with multi-scale targets: Traditional methods rely on fixed-sized convolution kernels or feature pyramids for perception. Such a static extraction strategy fails to capture the continuous scale variation from large nearby to small distant objects and severely underperforms in environments with extreme scale differences, like UAV surveillance. Recent methods enhance the feature of multimodal features by incorporating global interaction mechanism (such as cross-attention or Transformer modules) to enhance the fusion of multimodal features [7], [8], or by adopting target-aware strategies to reduce background interference [9]. However, such methods fail to preserve crucial shallow spatial details during deep fusion. This will have a negative impact on the performance in multi-scale scenarios, especially for distant targets.

On the other hand, the existing methods have limitations in effectively fusing complementary cross-modal information. Multispectral feature representation contains the private modality-specific information and the shared modality-consistent information. Blindly mixing the two without distinction will introduce the unique coding features of one modality (such as irrelevant background) will be introduced into the other mode. The modal information will be contaminated, the discriminative ability of the fusion features and the semantic consistency will be weakened, resulting in incorrect judgments as shown in Fig. 1(c) and Fig. 1(d). To address this challenge, it is crucial to explicitly process the two types of information, and to achieve the optimal balance between enhancing feature complementarity and maintaining the characteristics of each modality. Most existing fusion methods commonly employ a static weighting strategy that cannot adapt to individual instances, thus leading to excessive reliance on a dominant

This work was supported by the National Natural Science Foundation of China (No. 62576295), the Joint Fund of National Natural Science Foundation of China (No. U2468207), Chengdu Science and Technology Program (No. 2024-YF05-00340-SN), Sichuan Science and Technology Program (No. 2024ZDZX0001), and 2025 Southwest Jiaotong University International Student Teaching Reform and Education Management Research Project (No. 25LXSG14).

Fanyou Zhou and Jie Hu are with the School of Computing and Artificial Intelligence, Southwest Jiaotong University, Chengdu 611756, China (e-mail: jiehu@swjtu.edu.cn).

*Corresponding author

Bo Peng, Shengdong Du are with the School of Computing and Artificial Intelligence, Southwest Jiaotong University, Chengdu 611756, China.

Tianrui Li is with the School of Computing and Artificial Intelligence, Southwest Jiaotong University, Chengdu 611756, China, and also with the National Engineering Laboratory of Integrated Transportation Big Data Application Technology, Chengdu 610031, China.

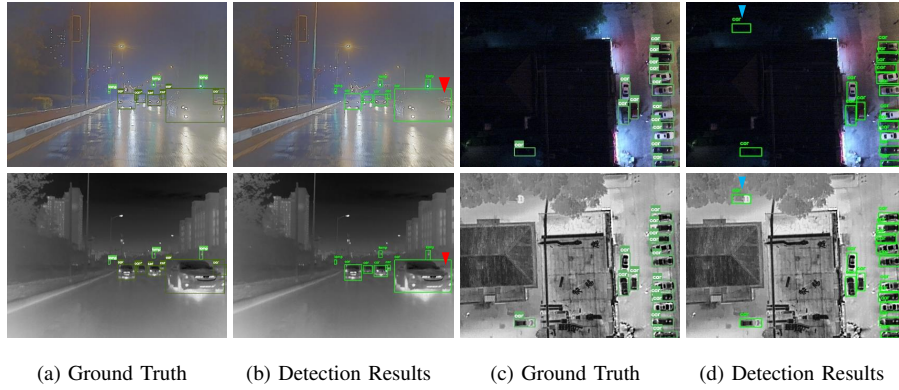


Fig. 1. Visualization of objects in paired RGB-thermal images. The first and second rows are RGB and thermal images respectively. In the two examples on the left, the model reliably identified the large-scale targets, but failed to capture the vehicle targets located further to the right of the picture, resulting in a false alarm situation (as indicated by the red arrow). In the two columns on the right, in the occluded area of the RGB image, the model mistakenly identified the thermal radiation contours in the infrared image scene as vehicle targets, thereby resulting in a false negative result (as indicated by the blue arrow).

modality [10], [11], [12]. Intuitively, the contribution of a low-quality modality (e.g., one corrupted by shadows or overexposure) should be dynamically suppressed. To address this, some studies have adopted perceptual weighting strategies [13], [14]. However, the dynamic weight estimation relies on an additional sub-network, introducing non-trivial computational overhead and complicating the training process. More importantly, the preset perception weights are not necessarily related to the multispectral targets, and thus cannot achieve accurate detection [15]. In addition, some general fusion frameworks have been proposed [16]. Nevertheless, since this framework is not specifically designed for the characteristics of a particular pattern, the performance in specialized tasks (e.g., detecting objects in thermal images) can be suboptimal.

To address above challenges, this paper proposes a multi-spectral object detection model named MPFNet. MPFNet consists of two core modules: the multi-scale perception convolution module (MPC) and the dynamic adaptive fusion module (DAF). Specifically, MPC is embedded within the main feature extraction network. Through the combination of convolution kernels of different scales, a continuously expanding perception domain is formed. Achieve efficient computing and more effective integration of multi-scale features, while preserving the local details of small targets. DAF is based on the adaptive cross-attention mechanism, dynamically allocating appropriate weights for the fusion of cross-modal features. It can achieve the fusion of complementary information of inter-modality while preserving the original characteristics of intra-modality. The two modules work together. The former provides rich and discriminative multi-scale feature representations, while the latter dynamically adjusts the weights to integrate cross-modal information. Finally, we introduce a Cross-Modal Contrastive Consistency (CMCC) loss. It constructs positive and negative samples within each modality to explicitly guide the fusion process, thereby enhancing the network's capacity to generate higher-quality fused features.

Our primary contributions can be summarized as follows:

- This paper presents a multi-scale perception fusion network named MPFNet for multispectral object detec-

tion. MPFNet effectively integrates multispectral features while enhancing the representation of multi-scale target regions.

- The proposed multi-scale perception convolution module (MPC) employs a cascaded depthwise separable convolution structure, which facilitates efficient multi-scale feature extraction and enhancement within a single forward pass.
- A dynamic adaptive fusion module (DAF) based on bidirectional cross-attention mechanism is designed, which employs bidirectional feature guidance between modalities to fully explore and integrate complementary information, effectively resulting in fused features with enhanced discriminability and semantic consistency.
- A cross-modal contrastive consistency (CMCC) loss is proposed for multispectral object detection, which operates through a positive-negative sample contrastive strategy to provide direct supervision for the fused features.

II. RELATED WORK

A. Multispectral object detection

Recent years have witnessed substantial advances in deep learning-based object detection through dual-modal fusion. The two-stream convolutional network has emerged as a predominant architecture in this domain. In this framework, images from distinct modalities are processed via separate yet parallel branches, followed by feature fusion at an intermediate stage, after which the consolidated features are forwarded to a detection head for final prediction. YOLO-Fusion [17] adapted the YOLO architecture into a dual-branch framework. It incorporates a Cross-Modal Adaptive Feature Fusion module (CMAFF) to integrate features from the two modalities in a divide-and-conquer manner. And a feature pyramid network similar to PANet was adopted to integrate multi-scale and multi-level features, thereby further refining spatial and semantic information. Similarly, LRAF-Net [18] employed a dual-stream CSPDarknet53 [19] network based on YOLO to improve detection performance by integrating the long-range dependencies of enhanced RGB and thermal

features. ICAFusion [7] designed a cross-modal cross-attention module (ICA) within the YOLOv5 [20] architecture. By constructing a novel calibrated complementary Transformer to address the issues of modality alignment errors and inaccurate fusion. Differing from the above, reference [21] proposed a YOLO-based detector characterized by a partially shared two-stream backbone. In this design, certain convolutional layers were shared across modalities to reduce parameters. While other layers remain independent to preserve the features representation of specific modalities.

The aforementioned studies have advanced the development of multispectral target detection technology. Simultaneously, the YOLO series has achieved an excellent balance between speed and accuracy by incorporating CSPNet, making it a widely adopted benchmark for real-time target detection tasks. Therefore, in this paper, we adopt the YOLO architecture as the network framework and identify that existing studies have yet to explore specialized approaches for handling targets at different scales, as well as methods for deeply mining dual-modal information to obtain reliable fusion features. To address this, two customized modules have been added to enhance the efficiency of multispectral target detection.

B. Multi-scale feature extraction

In multispectral object detection, the significant differences in target sizes present a common challenge. Therefore, multi-scale information is needed to enhance the feature representations of different objects. With the widespread application of FPN [22] in this field, many improved methods have emerged one after another. IPG-Net [23] aimed to address information imbalance by incorporating image pyramids into its backbone. ABFPN [24] suggests employing multiple parallel multi-scale dilated convolutions to capture contextual information, thereby improving the detection performance of small targets. In Mona [25], multi-scale deep separable convolution was employed to extract features under different receptive fields, thereby increasing the cognitive dimension and forming composite multi-scale cognitive features. MCANet [26] designed multi-scale cross-axis attention for targets with diverse scale transformations, embedding multi-scale features into axial attention. Similarly, FBRT-YOLO [27] focused on combining deep semantic information in the backbone network. By using multi-scale convolution kernels, it effectively alleviates the information extraction imbalance during feature extraction. However, the aforementioned methods still have limitations such as insufficient feature integration and reliance on sparse sampling, which can easily lead to the loss of details and make it difficult to achieve a complete perception of multi-scale targets.

We analyzed that integrating multi-scale features from different receptive fields is crucial, which enables the model to simultaneously capture fine-grained details and high-level semantics. Inspired by [27], we propose a multi-scale perceptual convolutional module. This module can capture multi-level features ranging from local details to the global background within a single module. Additionally, residual connections are added to effectively prevent information loss during the

transformation process, thereby ensuring the acquisition of rich and multi-scale feature representations even in complex scenarios.

C. Attention mechanism

The attention mechanism has been widely applied in computer vision tasks, and its effectiveness in improving model performance has been fully demonstrated. In 2021, FCA-Net [28] improved it by introducing the frequency domain into the attention mechanism, achieving progress in the field of image classification. The CMX [16] utilized cross-attention to achieve feature fusion between visible and auxiliary modalities. C2Former [29] developed an adaptive feature sampling method to capture complimentary characteristics. ICA-Fusion [7] introduced a iterative cross attention interaction approach. SimAM [30] developed an iterative cross-attention and parameter-sharing mechanism based on CBAM [31], which enhanced the exploration of the similarities between the two. GAFF [32] utilized segmentation masks as guidance to calculate spatial and channel attention weights within and between modalities, and performs adaptive features fusion via predicting the pedestrian masks. MSTF [33] introduced joint position encoding and enhanced cross attention. CARNet [34] presented a cross-attention feature reconstruction approach that flows information from strong to weak features across two modalities.

Although the aforementioned methods have enhanced the fusion performance through the attention mechanism [35], [15], they still have certain limitations. The traditional attention mechanism usually only focuses on enhancing the features of the same modality, without exploring the complementary information between different modalities, and lacks the adaptability to dynamically adjust the fusion strategy. Therefore, we proposes a dynamic architecture to interaction between modalities through a bidirectional cross-attention mechanism.

III. METHOD

A. Framework overview

The proposed MPFNet framework is shown in Fig. 2. Our network employs a dual-branch backbone network architecture comprising two primary components: a multi-scale perception convolution module (MPC) and a dynamic adaptive fusion module (DAF). We select YOLOv8 as the basic backbone and redesigned it into a dual-branch structure. It takes a pair of visible-light and infrared images as the input. In each branch, MPC module is used instead of the original convolution to extract features from the visible light and infrared images. Subsequently, DAF module fuses the dual-modal features. Finally, the fused features are input into the detector to obtain detection results.

B. Multi-scale perception convolution module

In the object detection task, the fixed kernel size of the convolution makes it difficult to simultaneously capture local details and global context information, thereby limiting the feature representation capability. When there are significant

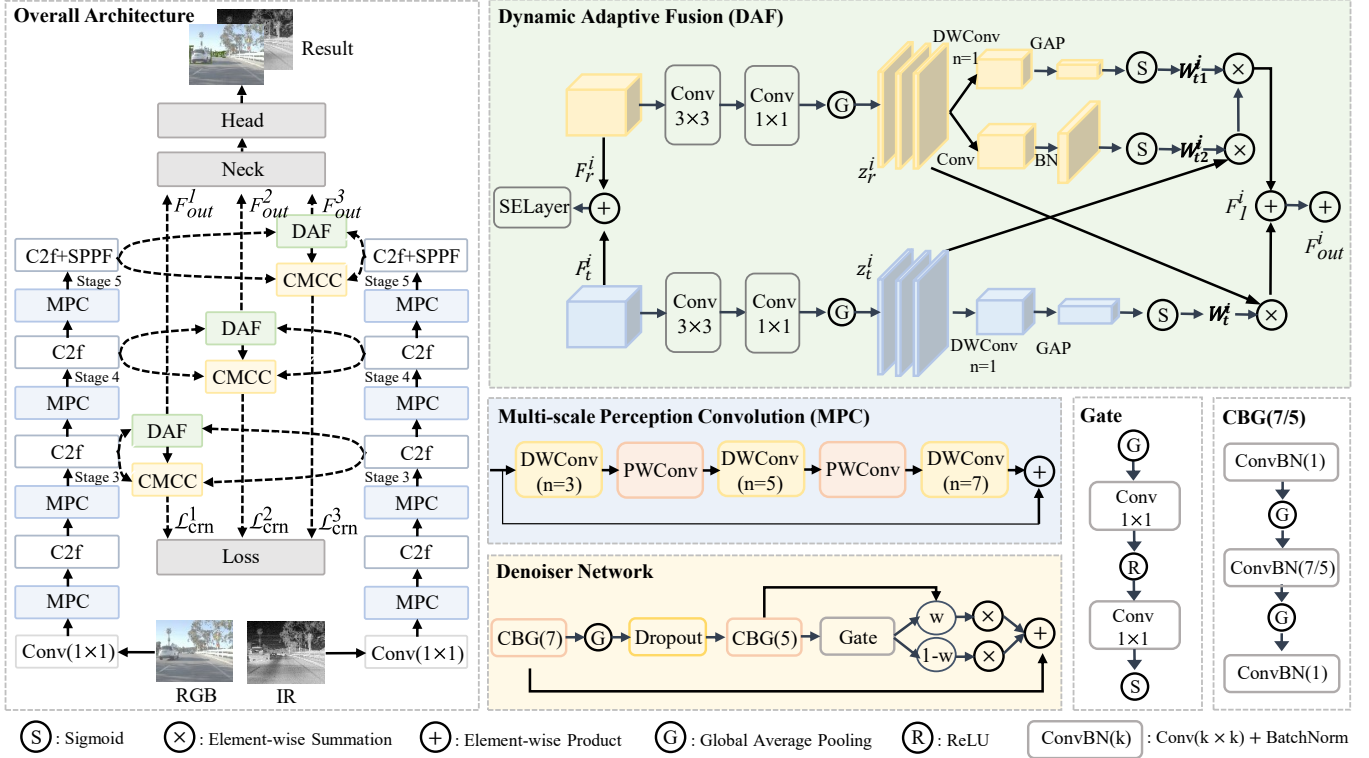


Fig. 2. The proposed MPFNet framework is summarized, consisting of two parallel branches of RGB and thermal modalities. MPC represents the multi-scale perception convolution module, DAF represents the dynamic adaptive fusion module, and CMCC represents the contrast consistency loss. The denoiser network is used for the construction of positive samples in CMCC.

differences in object scales, this problem becomes even more prominent. Therefore, with the core of achieving multi-scale feature extraction in a single layer to enhance the representation of weak targets, we design MPC to replace the convolutions in the original backbone (as shown in Fig. 2). And multi-scale global features f_R^i and f_T^i are obtained according to the following formulas:

$$f_R^i = \tau_{2n+1} (\mathcal{A} (\cdots \mathcal{A} (\tau_n (f_r^i)))) , \quad (1)$$

$$f_T^i = \tau_{2n+1} (\mathcal{A} (\cdots \mathcal{A} (\tau_n (f_t^i)))) , \quad (2)$$

where f_r^i and f_t^i represent the RGB and thermal feature of the i -th layer, respectively. τ_n indicates the use of a deep convolution with a kernel size of n , and we set n to 3. $\mathcal{A}$ represents point-wise convolution transformation. MPC employs a strategy of progressively expanding receptive fields. At each stage, depthwise and pointwise convolutions are combined: the former processes spatial information with significantly fewer parameters, while the latter handles channel-wise relationships, achieving preliminary feature enhancement. In addition, we add residual connections at the end of the module, connecting the features obtained from the above operations with the original input to ensure training stability. The final outputs F_r^i and F_t^i of the MPC module after residual connection at the i -th layer are obtained through the following calculations:

$$F_r^i = f_R^i + f_r^i, \quad (3)$$

$$F_t^i = f_T^i + f_t^i, \quad (4)$$

In this way, we utilize the MPC module to capture features of different scales, and achieve feature enhancement through progressive processing and residual connections.

C. Dynamic adaptive fusion module

Different imaging principles for RGB and thermal images result in varying physical parameters. Traditional methods typically rely heavily on the integrity of dual-modal information, resulting in a significant decline in detection performance if either modality fails. Therefore, many methods introduce pre-training fusion weights to improve accuracy. However, fixed weights cannot dynamically adjust to the constantly changing environment, especially in complex situations, where their performance becomes unstable. In addition, although the attention mechanism can be effective in the fusion of visible light and infrared images, compressing the features often results in a significant loss of information. Facing these challenges, we design DAF (as shown in Fig. 2). The RGB features and thermal features z_r^i and z_t^i of the i -th layer, which integrate global information, are calculated respectively by the following formulas (both have the same size):

$$z_r^i = \text{GAP} (\text{Conv}_{1 \times 1} (\text{Conv}_{3 \times 3} (F_r^i))) , \quad (5)$$

$$z_t^i = \text{GAP} (\text{Conv}_{1 \times 1} (\text{Conv}_{3 \times 3} (F_t^i))) , \quad (6)$$

where F_r^i and F_t^i respectively represent the multi-scale RGB and thermal features extracted from the i -th layer. $\text{Conv}_{1 \times 1}$ and $\text{Conv}_{3 \times 3}$ respectively represent convolutional layers with convolution kernels of size 1×1 and 3×3 . GAP indicates the

global average pooling operation. Specifically, the extracted features of the dual-modal approach are used as the input for DAF. Firstly, a 3×3 convolution layer is employed to extract local information. Then, a 1×1 convolution layer is used to unify the channels in terms of dimensionality, and pooling alignment is applied to adjust the spatial dimension, thereby obtaining a global description at the channel level. It is worth noting that the networks that obtain z_r^i and z_t^i are exactly the same, which is also more conducive to the subsequent information interaction and integration.

After obtaining high-quality single-modal features, achieving effective cross-modal fusion is the core issue. Generally speaking, traditional global pooling tends to lose spatial information. Therefore, it is usually combined with spatial attention mechanisms or other operations that preserve spatial structure. Unlike traditional simple fusion methods, we adopt a complementary mapping strategy between modalities, achieving intelligent fusion through a bidirectional cross-attention mechanism. In this approach, the two modalities provide mutual guidance, rather than being limited to one modality merely supplementing the other. The channel vectors and the spatial attention maps of the thermal features W_{t1}^i and W_{t2}^i are obtained through the following formulas:

$$W_{t1}^i = \sigma(\text{GAP}(\tau_3(z_r^i))), \quad (7)$$

$$W_{t2}^i = \sigma(\text{BN}(\text{Conv}_{1 \times 1}(z_r^i))), \quad (8)$$

where τ_3 indicates the use of a deep convolution with a kernel size of 3. σ represents the sigmoid activation function. BN represents normalization. The channel vector W_r^i of RGB is obtained through the following formula:

$$W_r^i = \sigma(\text{GAP}(\tau_3(z_t^i))), \quad (9)$$

The output F_1^i obtained after weighted processing at the i -th layer is obtained by the following formula:

$$F_1^i = F_r^i \cdot W_r^i + F_t^i \cdot W_{t1}^i \cdot W_{t2}^i, \quad (10)$$

RGB images contain rich spatial structure and spectral features. Through the channel attention and spatial attention mechanisms, more significant color targets can be captured. In contrast, thermal images have fewer texture details and mainly rely on contours formed by temperature variations for recognition. Therefore, in DAF, a cross-masking strategy is adopted. Dual-dimensional attention processing is applied to the visible modality, calibrating the thermal features from both the channel and spatial perspectives. The channel attention mechanism is applied to the thermal modality, focusing on the areas with significant changes in thermal radiation, and enhancing the RGB features from the channel perspective. A cyclic pattern is formed: the morphological characteristics of visible feature provide detailed information for the target location in the thermal modality, while the thermal image offers contour information for the visible modality. This asymmetric attention structure design is based on the different physical properties of RGB and thermal images, ensuring performance while achieving the optimal allocation of computing resources.

In addition, the fused features should not only explain the complementary relationship between the two modalities but

also retain their original characteristics. Therefore, the fused feature F_{out}^i is obtained by element-wise addition according to the following formula:

$$F_{out}^i = F_1^i + \text{SE}(F_r^i + F_t^i), \quad (11)$$

where F_r^i and F_t^i represent the RGB and thermal features of the input DAF at the i -th layer, respectively. SE indicates the SE layer [36]. $F_{out}^i \in \mathbb{R}^{C \times H \times W}$, has the same size as the input features. In this way, after adding the original features and performing SE processing on them, the effective information can be enhanced. Through the above operations, the fused features are finally obtained.

D. Loss function

The overall loss function of multispectral target detector typically comprises three components: bounding box regression loss ($\mathcal{L}_{\text{box}}$), classification loss ($\mathcal{L}_{\text{cls}}$), and confidence loss ($\mathcal{L}_{\text{conf}}$). However, such losses lack direct supervision of the fused features, which may continuously introduce noise during the fusion process. To reduce the failure of fusion caused by modality differences, we introduced a cross-modal contrastive consistency (CMCC) loss inspired by the InfoNCE loss [37]. The CMCC loss consists of two parts: positive constraints and negative constraints. The specific process of the positive sample constraint is as follows: Firstly, in the 3rd, 4th, and 5th stages of the network architecture shown in Fig. 2, lightweight convolutional network denoisers are constructed to extract positive samples. The dual-modal features $F^i \in \{F_r^i, F_t^i\}$ ($i \in \{3, 4, 5\}$) of this stage are used as the input for this process. CBG5/7 consists of consecutive GAP and ConvBN (as shown in Fig. 2). Each ConvBN consists of Conv and BatchNorm, where the numbers following convbn indicate the size of the convolution kernel. The former calculates spatial local information through sliding filters, while the latter performs normalization to suppress outliers. Subsequently, a gated network is designed to calculate the importance weights of the features, utilizing average pooling and two convolutions to determine the extent of retaining the original information (as shown in Fig. 2). Finally, the positive sample features P^i ($P^i \in \{P_r^i, P_t^i\}$) and the feature importance weights w are obtained in the i -th stage are calculated using the following formula:

$$m^i = \text{CBG5}(\text{Dropout}(\text{GAP}(\text{CBG7}(F^i)))), \quad (12)$$

$$w = \text{Gate}(m^i), \quad (13)$$

$$P^i = w \cdot m^i + (1 - w) \cdot \text{CBG7}(F^i), \quad (14)$$

where $m^i \in \{m_r^i, m_t^i\}$, represents the intermediate variable. The fused feature F_{out}^i ($i \in \{3, 4, 5\}$) of this layer serves as the anchor point for the samples. The InfoNCE value is calculated using the positive samples to narrow the gap between the positive samples and the anchor points. The positive constraint $\mathcal{L}_{\text{pos}}^i$ can be obtained by the following formula:

$$\mathcal{L}_{\text{pos}}^i = \log(\exp(\cos\text{Sim}(\frac{F_{out}^i, P_r^i}{\tau})) \quad (15)$$

$$+ \exp(\cos\text{Sim}(\frac{F_{out}^i, P_t^i}{\tau}))), \quad (16)$$

where $\cos\text{Sim}$ represents the cosine similarity. P_r^i and P_t^i respectively denote the denoised features obtained at layer i . τ represents the temperature coefficient, and we set its value to 0.3. If the value of τ is too small, it will lead to overfitting or unstable training; conversely, the model will have difficulty learning subtle differences. Therefore, a balanced value should be selected to strike a balance between amplifying the similarity differences and maintaining training stability. The average negative sample N^i and the negative constraint $\mathcal{L}_{\text{neg}}^i$ can be obtained by the following formula:, are calculated as follows:

$$N^i = \frac{n_r^i + n_t^i}{2}, \quad (17)$$

$$\mathcal{L}_{\text{neg}}^i = 1 - \log \left(1 - \exp \left(\cos\text{Sim} \left(\frac{F_{\text{out}}^i, N^i}{\tau} \right) \right) \right), \quad (18)$$

where n_r^i , n_t^i , respectively, denote the negative sample features obtained in the i -th layer. τ remains at 0.3.

After obtaining the positive sample constraints and the negative sample constraints, they are weighted and summed together to obtain the contrastive consistency loss for this stage. The calculation process is as follows:

$$\mathcal{L}_{\text{cnr}}^i = \alpha \cdot \mathcal{L}_{\text{pos}}^i + \beta \cdot \mathcal{L}_{\text{neg}}^i, \quad (19)$$

$$\mathcal{L}_{\text{cnr}} = \frac{1}{n} \sum_{i=1}^n \mathcal{L}_{\text{cnr}}^i, \quad (20)$$

where both α and β are set to 0.3. This causes the feature distributions of the two modalities to become more consistent, avoiding one-sided suppression. The effects of different weight combinations are presented in the ablation experiments in the Section IV. After repeating this operation in the 3rd and 4th stages, three stages of contrastive consistency losses are obtained. The average value is taken to get the final result, with n set to 3. Ultimately, the total loss function $\mathcal{L}_{\text{total}}$ is as shown in the following equation:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cnr}} + \mathcal{L}_{\text{box}} + \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{conf}}. \quad (21)$$

where $\mathcal{L}_{\text{box}}$ represents the bounding-box regression loss, $\mathcal{L}_{\text{cls}}$ represents the classification loss, and $\mathcal{L}_{\text{conf}}$ represents the confidence loss, which is consistent with the loss function obtained by the official detection head of YOLOv8.

IV. EXPERIMENTS

A. Datasets

We evaluate the performance of our model on three widely used public datasets: DroneVehicle [38], M3FD [39], and FLIR [40].

DroneVehicle dataset is a large public dataset of aerial vehicle images captured by drones, comprising 28,439 pairs of RGB and thermal images captured from day to night. The scenes cover various perspectives, heights, and lighting conditions in parking lots, urban roads, and residential areas. The dataset includes five object categories: “Car”, “Truck”, “Freight-car”, “Bus”, “Van”. The dataset is separated into training and test sets. Our experimental results are evaluated on the test set.

M3FD dataset is a multispectral object detection benchmark. Since the official of this dataset does not provide the data split, we use the split introduced in the work of TFDet [9]. 2905 pairs of images are used for training, and 1295 pairs of images are used for testing. We use the image pairs obtained by adjusting and padding the short sides of the dataset images due to the different resolutions of the images in this work. This dataset contains six image categories: “Person”, “Car”, “Bus”, “Motorcycle”, “Lamp”, and “Truck”.

FLIR Dataset is a widely used benchmark dataset in the field of RGB-thermal image fusion for object detection, covering challenging day-night transition scenarios. Initially, there was a problem of unaligned dual-modal images in this dataset. Subsequent studies corrected this issue and provided versions with strictly aligned image pairs. Therefore, we use the aligned dataset [41], which contains 5142 pairs of well-aligned RGB-thermal images, of which 4129 pairs are

used for training and 1013 pairs for testing. This dataset contains three object categories: “Person”, “Car”, and “Bicycle”.

Fig. 3 presents representative sample examples from the three datasets used in this paper. In the aerial images of DroneVehicle, situations where targets are densely arranged can be observed. In M3FD, representative samples with significant differences in target sizes are presented. In FLIR, a group of typical low-light night images is selected as examples. The above samples cover key challenges such as dense targets, multi-scale variations, and low light conditions at night, providing a data basis for a comprehensive assessment of the performance of this method.

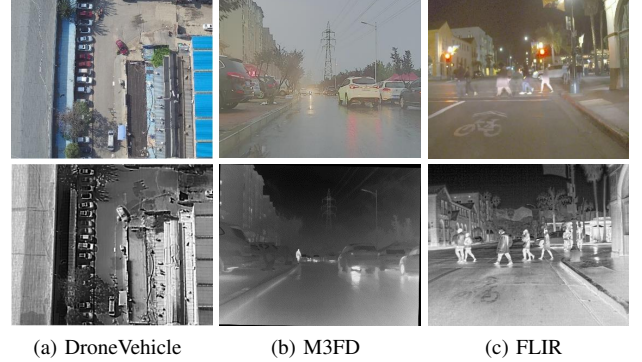


Fig. 3. Image examples from DroneVehicle, M3FD, and FLIR datasets. (The three groups of images have been adjusted to the same size for better appearance)

B. Experiment settings

All experiments in this paper adopt the YOLOv8 detector and are implemented using the MMDetection toolbox. Training is conducted on an NVIDIA A40 GPU, with the environment built on PyTorch 1.8.1 and CUDA 12.6. The training stage is set to 300 epochs, with a total batch size of 8 and a base learning rate of 0.01. The SGD optimizer is used with an initial learning rate of 1.0×10^{-2} and a momentum of 0.937. Input images are resized to 640×640 during training. Hyperparameter settings for the YOLOv8 detector remain consistent with those in the official configuration file.

C. Evaluation on DroneVehicle dataset

In order to systematically evaluate the performance of the proposed method on DroneVehicle dataset, we compared it with a variety of representative methods, including detection methods that are based solely on a single modality: RetinaNet [42], RoITransformer [43], R3Det [44], Oriented R-CNN [45], and PKINet-S [46]. In addition, there are the classic cross-modal detection methods, such as Halfway Fusion [47], as well as the recently proposed S2A-Net [48], TSFADet [49], TarDAL [39], C2Former [29], and CAGTDet [10]. Table I presents the comparison results, from which it can be observed that the proposed method has achieved the best performance.

Specifically, in the detection method using mono-modality RGB, RoITransformer [43] achieved the best performance. In the detection method using the IR modality, PKINet-S [46] achieved the best performance. Although the detection performance of some categories in the dual-modal approach is inferior to that of the single-modal approach, overall, the use of dual-modal has significantly improved the overall detection accuracy. This confirms the complementarity and effectiveness of multi-spectral target detection. The proposed method achieved the best performance in the bimodal experiment. Our analysis indicates that the performance improvement is primarily attributed to the synergistic effect of the design architecture proposed in this paper. MPC effectively captures the cross-scale information from local details to the global context. On this basis, DAF fully leveraged its complementary advantages to achieve a deep level of

TABLE I
DETECTION RESULTS ON DRONEVEHICLE DATASET (MAP, IN %). THE BEST RESULT IS HIGHLIGHTED IN BOLD.

Methods	Modality	Car	Truck	Freight-car	Bus	Van	mAP ₅₀	mAP
RetinaNet [42]	RGB	78.5	34.4	24.1	69.8	28.8	25.0	23.4
RoITransformer [43]		61.6	55.1	42.3	85.5	44.8	61.6	32.9
R3Det [44]		80.3	56.1	42.7	80.2	44.4	27.4	27.4
Oriented R-CNN [45]		80.1	53.8	41.6	85.4	43.3	60.8	32.7
RetinaNet [42]	thermal	88.8	35.4	39.5	76.5	32.1	54.5	30.4
R3Det [44]		89.5	48.3	16.6	87.1	39.9	62.3	36.7
Oriented R-CNN [45]		89.6	53.9	53.9	89.2	41.0	66.0	-
PKINet-S [46]		90.2	67.3	56.3	88.8	51.2	70.8	43.8
Halfway Fusion [47]	RGB+thermal	90.1	62.3	58.5	89.1	49.8	70.0	-
S2A-Net [48]		90.0	64.5	61.7	88.2	53.2	71.5	-
TSFADet [49]		89.9	67.9	63.7	89.8	54.0	73.1	-
TarDAL [39]		89.5	68.3	56.1	89.4	59.3	72.6	43.3
C2Former [29]		90.2	68.3	64.4	89.8	58.5	74.2	47.5
CAGTDet [10]		90.8	70.0	66.3	90.5	55.6	74.5	-
MPFNet(Ours)		96.2	72.7	63.2	94.3	63.2	78.7	57.3

integration. Although in the “Freight-car” category, our method is slightly lower by 3.1 % compared to CAGTDet [10], the detection results in the other categories were significantly better than the comparison method, and achieved an advantage in overall accuracy. This verifies that the proposed method has more stable detection capabilities in the vast majority of scenarios.

D. Evaluation on M3FD dataset

On M3FD dataset, we compared our method with DIDFuse [50], TarDAL [39], RFNet [51], CFT [52], ICAFusion [7], CDDFusion [53], and MFFNet [54]. The results are shown in Table II.

TABLE II
DETECTION RESULTS ON M3FD DATASET (MAP, IN %). THE BEST RESULT IS HIGHLIGHTED IN BOLD.

Methods	mAP ₅₀	mAP ₇₅	mAP
DIDFuse [50]	78.9	-	52.6
TarDAL [39]	80.4	-	53.1
RFNet [51]	80.4	-	53.2
CFT [52]	85.3	44.6	57.1
ICAFusion [7]	78.2	44.5	40.4
CDDFusion [53]	81.1	-	54.3
MFFNet [54]	88.1	-	57.6
MPFNet(Ours)	88.4	63.3	59.3

The results show that our method is highly competitive. Specifically, MPFNet outperforms TarDAL [39] by 6.2 % in mAP and has achieved a 1.7 % improvement over the second-best model, MFFNet [54], which is sufficient to demonstrate that the proposed method exhibits superior overall performance. In particular, our model outperformed the methods CFT [52], ICAFusion [7] and MFFNet [54] that using the same CSPDarknet53v5 backbone. The results on M3FD dataset demonstrate that our model has strong detection capabilities. DAF module handles the fusion of different modalities more comprehensively and outperforms other fusion methods.

E. Evaluation on FLIR dataset

On the Flir dataset, we compared our method with six other approaches that employed different models, including the classic models Yolov5s [20], Yolov11s [55], CFT [52], as well as recently proposed methods ICAFusion [7], LRAF-Net [18], and RSDet [56]. The comparison results are shown in Table III. The proposed method outperforms many advanced methods, with an mAP that is 3.6 % higher than that of the second-best model, RSDet [56]. Although

the mAP₅₀ metric is slightly lower than that of RSDet [56], the method presented in this paper demonstrates significant advantages under more stringent evaluation criteria. Overall, this method is more competitive in scenarios that require high precision localization.

TABLE III
DETECTION RESULTS ON FLIR DATASET (MAP, IN %). THE BEST RESULT IS HIGHLIGHTED IN BOLD.

Methods	mAP ₅₀	mAP ₇₅	mAP
Yolov5s [20]	72.8	32.0	37.1
Yolov11s [55]	74.6	33.8	39.6
CFT [52]	77.1	35.2	39.8
ICAFusion [7]	78.2	36.5	40.4
LRAF-Net [18]	79.5	-	40.8
RSDet [56]	80.1	-	40.1
MPFNet(Ours)	78.6	38.3	43.7

F. Ablation studies

To verify the effectiveness of each component of the proposed method, the following ablation experiment results are presented for MPC, DAF, and CMCC loss. Unless otherwise specified, the experimental groups that do not include the DAF module all use the feature addition method for fusion.

Effectiveness of MPC module. We conduct ablation experiments for the MPC module on three datasets. The experimental results are recorded in Tables IV to VI. As shown in IV, the overall detection performance of the model has improved on DroneVehicle dataset after the introduction of MPC. The feature extraction layer uses MPC module to effectively enhance the multi-scale information representation capability through progressive receptive fields. Moreover, the detection accuracy of each category has been consistently improved, further verifying the MPC module’s powerful ability to capture features at multiple levels, from local details to global context. The experimental results on M3FD and FLIR datasets, as shown in Tables V and VI. Obviously, the detection accuracy has significantly improved. This indicates that in different scenarios, the module enhances the ability to preserve local details, thereby achieving a rich representation of multi-scale features and effectively improving the distinction between the target and the background. Therefore, the effect of MPC module in enhancing detection performance in various situations is remarkable.

Effectiveness of DAF module: To assess the contribution of the DAF module, we conduct tests on different datasets, and the results are shown in Tables IV and V. In Table IV, the third set of experiments

TABLE IV
ABLATION ON DRONEVEHICLE DATASET (MAP, IN %). THE BEST RESULT IS HIGHLIGHTED IN BOLD.

ID	MPC	DAF	CMCC	Car	Truck	Freight-car	Bus	Van	mAP ₅₀	mAP
1				90.1	67.3	62.4	87.5	59.4	73.5	49.7
2	✓			92.1	70.2	63.1	89.9	60.7	75.2	54.1
3		✓		92.2	70.3	64.5	90.0	60.8	75.6	54.8
4	✓	✓		94.3	70.4	65.4	92.9	62.8	77.2	55.3
5		✓	✓	95.1	69.0	64.6	93.6	60.3	76.5	55.5
6			✓	93.3	68.7	63.9	90.2	61.2	75.5	55.3
7	✓	✓	✓	96.2	72.7	63.2	94.3	63.2	78.7	57.3

TABLE V
ABLATION ON M3FD DATASET (MAP, IN %). THE BEST RESULT IS HIGHLIGHTED IN BOLD.

ID	MPC	DAF	CMCC	mAP ₅₀	mAP ₇₅	mAP
1				84.2	59.9	58.1
2	✓			85.7	61.9	58.4
3		✓		86.8	62.6	58.6
4	✓	✓		87.6	62.6	58.7
5		✓	✓	87.5	62.3	58.8
6			✓	86.6	61.9	58.7
7	✓	✓	✓	88.4	63.3	59.3

TABLE VI
ABLATION ON FLIR DATASET (MAP, IN %). THE BEST RESULT IS HIGHLIGHTED IN BOLD.

Methods	mAP ₅₀	mAP ₇₅	mAP
Baseline	77.8	37.2	40.6
+MPC	78.3	40.5	43.1

clearly demonstrates that after using the DAF module alone, the mAP₅₀ increased from the baseline of 73.5% to 75.6%, and the detection accuracy of each category is significantly has improved. This indicates that DAF is capable of dynamically integrating discriminative features from each modality, thereby enhancing the perception ability of the model in the fused representation. Furthermore, we evaluate the additional value of DAF by incorporating the MPC module. As shown in the second and fourth sets of experiments in Table V, the addition of DAF improved the overall performance of the model, indicating that this module can complement the multi-scale feature extraction mechanism while demonstrates its effectiveness under different baselines. Additionally, the experimental results on M3FD dataset support the above conclusion, as shown in Table V. Whether or not MPC is introduced, DAF always brings stable performance improvements, which indicates that DAF has strong adaptability in different architectures.

To further verify the effectiveness of the DAF, we design three additional classic fusion strategies to be compared on DroneVehicle dataset: direct feature summation, normal convolution module, and feature connection. The results are shown in Table VII. Obviously, DAF outperforms other methods, which further verifies the effectiveness of DAF module in fusing feature maps.

Furthermore, in the process of obtaining z_r^i and z_t^i through pooling in the DAF module, we conduct experiments to verify that this is the most effective method for obtaining the initial global description, as shown in Table VIII. We replace pooling with the interpolation alignment method to adjust the spatial dimension. Although the mAP₅₀ of both methods are the same, the average pooling method was slightly higher than the interpolation alignment method in multiple categories and mAP. Therefore, we use the average pooling method. It is worth noting that neither of these two sets of experiments added the MPC and CMCC, only modifying the method of adjusting the spatial

TABLE VII
ABLATION STUDY OF DAF ON DRONEVEHICLE DATASET (MAP, IN %). THE BEST RESULT IS HIGHLIGHTED IN BOLD.

Methods	mAP ₅₀	mAP ₇₅	mAP
Add	76.7	66.5	55.2
Conv	76.8	66.0	54.8
Concat	76.9	66.6	55.3
DAF	78.7	69.1	57.3

dimension in the DAF module. Additionally, to avoid the randomness of the results, we conduct an ablation study on the feature alignment scheme on M3FD dataset under the configuration of simultaneously using MPC, DAF, and CMCC loss. The results are shown in Table VIII, shows that the best performance was achieved when using average pooling.

Effectiveness of CMCC loss: We conduct systematic ablation experiments to verify the effectiveness of the CMCC loss. Since the CMCC loss takes cross-modal features and fused features as inputs, in the experimental group without using the DAF module, we employ a method that directly adds features to fuse them. The results are shown in Table tables IV and V, in the sixth group of experiments. It can be clearly seen that even without introducing the MPC and DAF modules, the CMCC loss can improve the quality of the generated fused features through direct supervision, and the detection accuracy has significantly improved compared to the baseline first set of experiments. In addition, we conduct a combined ablation experiment between CMCC and DAF, as shown in the fifth group of experiments in Tables tables IV and V. When DAF and CMCC are used together, they exhibit a significant synergistic enhancement effect. DAF generates fused features rich in cross-modal information, while CMCC provides clear result-oriented supervision signals for it. The two modules together enhance the model's ability to capture spatial structures.

To further evaluate the effectiveness of the CMCC loss, we replace it with the widely used SSIM loss function in image tasks. The experimental results on M3FD dataset are shown in Table IX. CMCC achieved the best results in multiple key indicators. We concluded that CMCC leads to the generation of higher-quality fused features. Therefore, the model's ability to learn a reasonable structure in the learning space has been enhanced.

Additionally, to investigate the impact of different combinations of α and β on performance, we conduct three sets of comparative experiments using different weight schemes, and the results are presented in Table X. Since our method places greater emphasis on the relationship between the interfering information and the fused features, we all adopted the weight setting of $\alpha \leq \beta$. The results in the table clearly show that the weight configuration has a significant impact on the detection performance: when the value of α is too low, the model does not learn the cross-modal semantic consistency adequately. While if β is too high, it may lead to excessive denoising, weakening the subtle features beneficial for the detection task. When both α and β are set to 0.3, the model achieves a balance between retaining discriminative features and suppressing noise, thereby yielding the

TABLE VIII
ABLATION STUDY OF FEATURE ALIGNMENT SCHEMES ON DRONEVEHICLE AND M3FD DATASET (MAP, IN %). THE BEST RESULT IS HIGHLIGHTED IN BOLD.

Methods	Car	Truck	DroneVehicle			mAP ₅₀	mAP	mAP ₅₀	M3FD	
			Freight-car	Bus	Van				mAP ₇₅	mAP
Pooling Alignment	92.2	70.3	64.5	90.0	60.8	75.6	54.8	87.6	63.8	59.3
Interpolate Alignment	91.3	69.2	66.3	90.4	60.8	75.6	53.6	87.0	63.7	59.1

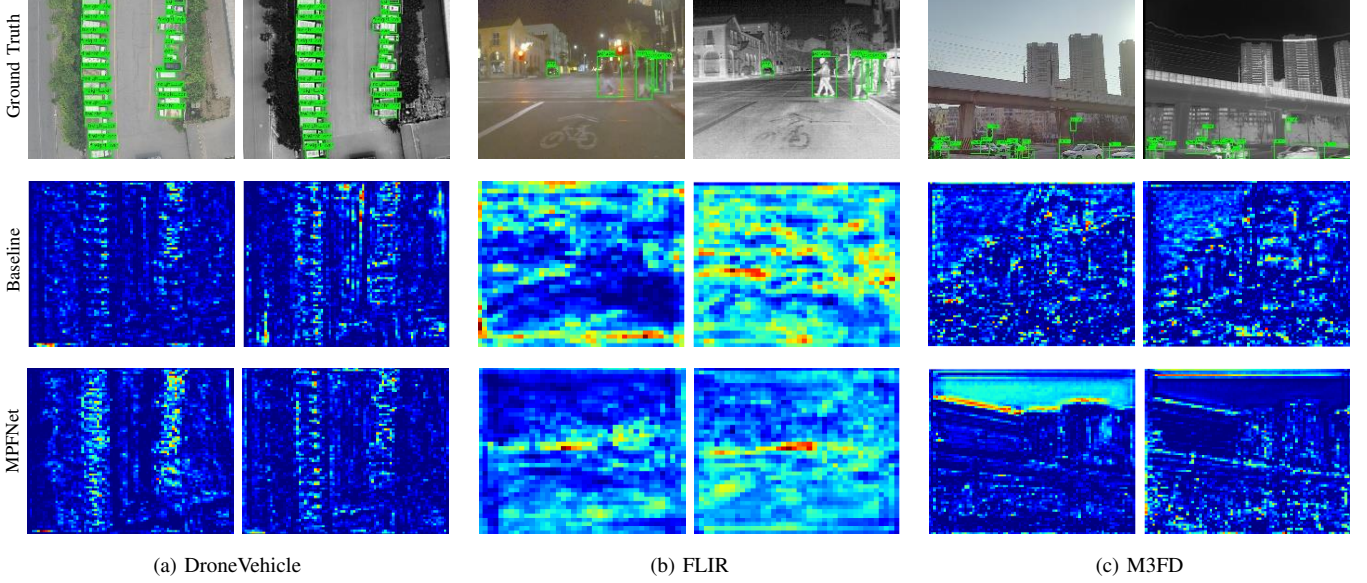


Fig. 4. Visualization results of feature maps on DroneVehicle, FLIR and M3FD datasets. For aesthetic display, we cropped and uniformly processed images of different sizes.

TABLE IX
ABLATION STUDY ON THE LOSS FUNCTION ON M3FD DATASET (MAP, IN %). THE BEST RESULT IS HIGHLIGHTED IN BOLD.

Methods	mAP ₅₀	mAP ₇₅	mAP
SSIM	87.5	63.4	59.1
CMCC	87.6	63.8	59.3

TABLE X
ABLATION STUDY OF WEIGHT COMBINATION SCHEMES ON DRONEVEHICLE DATASET (MAP, IN %). THE BEST RESULT IS HIGHLIGHTED IN BOLD.

α	β	mAP ₅₀	mAP ₇₅	mAP
0.2	0.5	78.0	67.8	56.1
0.3	0.3	78.7	69.1	57.3
0.1	0.6	77.1	67.7	55.6

best detection performance.

G. Visualized analysis

Features visualisation. Fig. 4 compares the feature attention maps between the baseline method and the method proposed in this paper on three datasets. The feature maps of both the baseline method and the proposed method are the features after dual-modal fusion. To ensure a fair comparison, all the feature maps are from the same output channel after fusion. It can be observed that, compared with the baseline model, the feature maps of the proposed method provide a clearer and more focused positioning of the target area. As shown

in the example of the DroneVehicle dataset in Fig. 4, the feature activations of the baseline model are relatively diffuse and difficult to identify the key information, while our method provides more accurate identification of the target area, with the interest region concentrated near the ground truth. Taking M3FD dataset as an example, the proposed method’s internal attention differs markedly from the baseline, which shows no focus on the target. Our approach distinctly learns object boundaries and concentrates on semantically dense regions (e.g., the lower image section). It indicates that MPFNet can effectively suppress the activation of interfering features in complex environments and enhance the signals of relevant targets.

Detection result. In the qualitative analysis, Fig. 5 presents the results of comparing our method with CFT [52] and ICAFusion [7] in terms of detection performance on FLIR and M3FD datasets. As shown in the figure, our method demonstrates excellent performance. Specifically, in scenarios with uneven nighttime lighting (Fig. 5 first and second columns), the method proposed in this paper successfully detected more targets and significantly reduced missed detections. In scenarios with cluttered backgrounds and small target sizes (third and fourth columns), the comparison method made different errors, while the method proposed in this paper effectively avoided such issues. In daytime scenarios with large differences in target sizes and dense distributions (fifth and sixth columns), the method proposed in this paper still demonstrated significant advantages. In MPFNet, MPC effectively captures features of different scales, enabling small targets to exhibit strong performance. Meanwhile, DAF further enhances the complementarity across modalities and the stability of the model, thereby making the distinction between the target and the background more significant. Although there may be occasional misjudgments and missed detections, our method has a competitive advantage in detecting in complex environments.

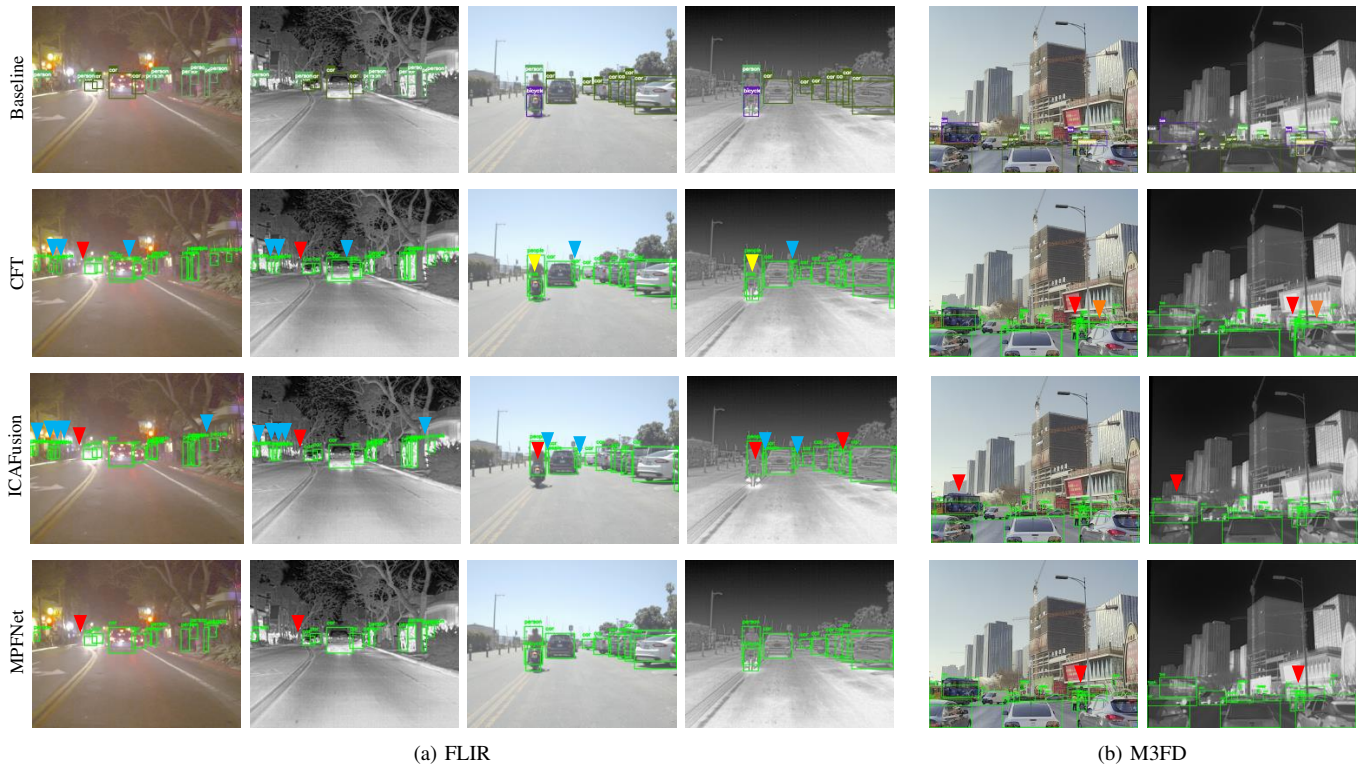


Fig. 5. Visual detection results generated by various methods from FLIR and M3FD datasets. The examples in the left four columns are selected from FLIR dataset. For examples in FLIR (as shown in (a)), the comparison methods show multiple false negatives (as indicated by the blue arrows), false positives (as indicated by the red arrows), and incorrect classifications (as indicated by the yellow arrows). The proposed method effectively avoids these errors. For the examples in M3FD (as shown in (b)), the comparison methods still show false positives and duplicate labels (as shown by the orange arrows). Our method is superior in the case. For aesthetic display, we cropped and uniformly processed images of different sizes.

V. CONCLUSION AND FUTURE WORK

This paper investigates key limitations in multispectral target detection, notably insufficient feature extraction, inefficient modality fusion, and weak supervisory signals. To address these issues, we have proposed a multi-spectral fusion detection network named MFD-Net based on YOLO. Our multi-scale perception convolution module (MPC) effectively extracts and retains multi-scale local detail features through progressive convolution, significantly enhancing the model’s ability to represent small and multi-scale targets. Furthermore, a dynamic adaptive fusion module (DAF) introduces a bidirectional cross-attention mechanism, which adaptively achieves feature complementarity and weight allocation between RGB and thermal images, thereby improving the robustness of dual-modal collaborative perception. In addition, a cross-modal contrastive consistency (CMCC) loss constructs contrastive supervision by matching positive samples in the target region with negative samples in the background region, directly guiding the alignment of fused features with target semantics and reducing background interference, which ensures the discriminative capability of the fused features. Our approach systematically addresses issues such as the loss of multi-scale details, insufficient utilization of modal complementarity, and weak discriminative power of fused features. Comprehensive experiments on multiple public datasets have shown that the proposed method has achieved superior performance. Nevertheless, our work admits avenues for further refinement. In the future, we will primarily focus on research in model lightweighting and deployment, exploring more efficient fusion architectures and compression techniques. At the same time, we plan to enhance the model’s practical performance in extreme situations by introducing more refined feature enhancement mechanisms and generalization strategies.

REFERENCES

- [1] Cao, Pang, Xie, Khan, and Shao, “From handcrafted to deep features for pedestrian detection: A survey,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 9, pp. 4913–4934, 2022.
- [2] Feng and Su, “Learning multi-layer attention aggregation siamese network for robust RGBT tracking,” *IEEE Transactions on Multimedia*, vol. 26, pp. 3378–3391, 2024.
- [3] Wu, Liu, Shi, Liu, and Li, “Style-agnostic representation learning for visible-infrared person re-identification,” *IEEE Transactions on Multimedia*, vol. 26, pp. 2263–2275, 2024.
- [4] Wang, Sun, Chi, and Shen, “RGB-T object detection with failure scenarios,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 18, pp. 3000–3010, 2025.
- [5] Wu, Jia, Chang, Wang, and Chen, “RGB-T saliency detection based on multiscale modal reasoning interaction,” *IEEE Transactions on Instrumentation and Measurement*, vol. 73, pp. 1–15, 2024.
- [6] Li, Liang, Lu, Zhao, and Tang, “RGB-T object tracking: Benchmark and baseline,” *Pattern Recognition*, vol. 96, p. 106977, 2019.
- [7] Shen, Chen, Liu, Zuo, Fan, and Yang, “ICAFusion: iterative cross-attention guided feature fusion for multispectral object detection,” *Pattern Recognition*, vol. 145, p. 109913, 2024.
- [8] Lee, Jovanov, and Philips, “Cross-modality attention and multimodal fusion transformer for pedestrian detection,” in *Proceedings of the Lecture Notes in Computer Science*, 2023, pp. 608–623.
- [9] Zhang, Zhang, Wang, Ying, Sheng, Yu, Li, and Shen, “TFDet: Target-aware fusion for RGB-T pedestrian detection,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 36, no. 7, pp. 13 276–13 290, 2025.
- [10] Yuan, Shi, Wang, Wang, and Wei, “Improving RGB-infrared object detection with cascade alignment-guided transformer,” *Information Fusion*, vol. 105, p. 102246, 2024.
- [11] Xie, Zhang, Yu, and Xie, “YOLO-MS: Multispectral object detection via feature interaction and self-attention guided fusion,” *IEEE Transactions on Cognitive and Developmental Systems*, vol. 15, no. 4, pp. 2132–2143, 2023.

- [12] Zhang, Yu, He, Wang, and Yang, "Illumination-guided RGBT object detection with inter- and intra-modality fusion," *IEEE Transactions on Instrumentation and Measurement*, vol. 72, pp. 1–13, 2023.
- [13] Li, Song, Tong, and Tang, "Illumination-aware Faster R-CNN for robust multispectral pedestrian detection," *Pattern Recognition*, vol. 85, pp. 161–171, 2019.
- [14] Kim, Park, and Ro, "Uncertainty-guided cross-modal learning for robust multispectral pedestrian detection," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 32, no. 3, pp. 1510–1523, 2022.
- [15] Li, Zhang, Hu, Zhu, Fu, and Chen, "Stabilizing multispectral pedestrian detection with evidential hybrid fusion," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 34, no. 4, pp. 3017–3029, 2024.
- [16] Zhang, Liu, Yang, Hu, Liu, and Stiefelhofen, "CMX: Cross-modal fusion for RGB-X semantic segmentation with transformers," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 12, pp. 14 679–14 694, 2023.
- [17] Qingyun and Zhaokui, "Cross-modality attentive feature fusion for object detection in multispectral remote sensing imagery," *Pattern Recognition*, vol. 130, p. 108786, 2022.
- [18] Fu, Wang, Duan, Xiao, Dian, Li, and Li, "LRAF-Net: Long-range attention fusion network for visible–infrared object detection," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 10, pp. 13 232–13 245, 2024.
- [19] Wang, Liao, Wu, Chen, Hsieh, and Yeh, "CSPNet: A new backbone that can enhance learning capability of CNN," in *Proceedings of the 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, jun 14–19, 2020.
- [20] Ultralytics, "YOLOv5 by Ultralytics," <https://github.com/ultralytics/yolov5>, 2023.
- [21] Zou, Peng, and Zhou, "UAV-Based human detection with visible-thermal fused YOLOv5 network," *IEEE Transactions on Industrial Informatics*, vol. 20, no. 3, pp. 3814–3823, 2024.
- [22] Lin, Dollár, Girshick, He, Hariharan, and Belongie, "Feature pyramid networks for object detection," in *Proceedings of the 2017 IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 936–944.
- [23] Liu, Gao, Sun, and Fang, "IPG-Net: Image pyramid guidance network for small object detection," in *Proceedings of the 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2020, pp. 4422–4430.
- [24] Zeng, Wu, Wang, Li, Liu, and Liu, "A small-sized object detection oriented multi-scale feature fusion approach with application to defect detection," *IEEE Transactions on Instrumentation and Measurement*, vol. 71, pp. 1–14, 2022.
- [25] Yin, Hu, Li, Zhang, and Yang, "5 % > 100 %: Breaking performance shackles of full fine-tuning on visual recognition tasks," in *Proceedings of the 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 20 071–20 081.
- [26] Shao, Zeng, Hou, and Yang, "MCANet: Medical image segmentation with multi-scale cross-axis attention," *Machine Intelligence Research*, vol. 22, no. 3, pp. 437–451, 2025.
- [27] Xiao, Xu, Xin, and Li, "FBRT-YOLO: Faster and better for real-time aerial image detection," in *Proceedings of the Thirty-ninth AAAI Conference on Artificial Intelligence, AAAI-25, VOL 39 NO 8*, ser. AAAI Conference on Artificial Intelligence, Walsh, Shah, and Kolter, Eds., 2025, pp. 8673–8681.
- [28] Qin, Zhang, Wu, and Li, "FCANet: frequency channel attention networks," in *Proceedings of the IEEE International Conference on Computer Vision*, 2021, pp. 763–772.
- [29] Yuan and Wei, "C2Former: Calibrated and complementary transformer for rgb-infrared object detection," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–12, 2024.
- [30] Yang, Zhang, Li, and Xie, "SimAM: A simple, parameter-free attention module for convolutional neural networks," in *Proceedings of the Machine Learning Research*, vol. 139, 2021, pp. 11 863–11 874.
- [31] Woo, Park, Lee, and Kweon, "CBAM: Convolutional block attention module," in *Proceedings of the European Conference on Computer Vision - ECCV 2018*, ser. Lecture Notes in Computer Science, vol. 11211, 2018, pp. 3–19.
- [32] Zhang, Fromont, Lefevre, and Avignon, "Guided attentive feature fusion for multispectral pedestrian detection," in *Proceedings of the 2021 IEEE Winter Conference on Applications of Computer Vision, WACV 2021*, 2021, pp. 72–80.
- [33] Hou, Yang, Zheng, and Gao, "Multispectral transformer fusion via exploiting similarity and complementarity for robust pedestrian detection," *Pattern Recognition*, vol. 162, p. 111383, 2025.
- [34] Wang, Sun, Dong, and Zheng, "CARNet: Cross-Attention guided feature reconstruction for RGB-T Object detection," *Expert Systems With Applications*, vol. 298, p. 129865, 2026.
- [35] Zhao and Zhang, "A feature divide-and-conquer network for RGB-T semantic segmentation," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 33, no. 6, pp. 2892–2905, 2023.
- [36] Hu, Shen, and Sun, "Squeeze-and-excitation networks," in *Proceedings of the 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2018, pp. 7132–7141.
- [37] Van Den Oord, Li, and Vinyals, "Representation learning with contrastive predictive coding," *arXiv*, 2018.
- [38] Sun, Cao, Zhu, and Hu, "Drone-based RGB-infrared cross-modality vehicle detection via uncertainty-aware learning," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 32, no. 10, pp. 6700–6713, 2022.
- [39] Liu, Fan, Huang, Wu, Liu, Zhong, and Luo, "Target-aware dual adversarial learning and a multi-scenario multi-modality benchmark to fuse infrared and visible for object detection," in *Proceedings of the 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 5792–5801.
- [40] F. Team, "Free teledyne FLIR thermal dataset for algorithm training," 2025. [Online]. Available: <https://oem.flir.com/solutions/automotive/adas-dataset-form/>
- [41] Zhang, Fromont, Lefevre, and Avignon, "Multispectral fusion for object detection with cyclic fuse-and-refine blocks," in *Proceedings of the International Conference on Image Processing, ICIP*, vol. 2020-October, 2020, pp. 276–280.
- [42] Lin, Goyal, Girshick, He, and Dollár, "Focal loss for dense object detection," in *Proceedings of the IEEE International Conference on Computer Vision*, vol. 2017-October, 2017, pp. 2999–3007.
- [43] Ding, Xue, Long, Xia, and Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, vol. 2019-June, 2019, pp. 2844–2853.
- [44] Yang, Yan, Feng, and He, "R3Det: Refined single-stage detector with feature refinement for rotating object," in *Proceedings of the 35th AAAI Conference on Artificial Intelligence, AAAI 2021*, vol. 4A, 2021, pp. 3163–3171.
- [45] Xie, Cheng, Wang, Yao, and Han, "Oriented R-CNN for object detection," in *Proceedings of the 2021 IEEE International Conference on Computer Vision*, 2021, pp. 3500–3509.
- [46] Cai, Lai, Wang, Wang, Sun, and Yao, "Poly kernel inception network for remote sensing detection," in *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, 2024, pp. 27 706–27 716.
- [47] Liu, Zhang, Wang, and Metaxas, "Multispectral deep neural networks for pedestrian detection," in *Proceedings of the British Machine Vision Conference, BMVC 2016*, 2016, pp. 73.1–73.13.
- [48] Han, Ding, Li, and Xia, "Align deep features for oriented object detection," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–11, 2022.
- [49] Yuan, Wang, and Wei, "Translation, scale and rotation: Cross-modal alignment meets RGB-infrared vehicle detection," in *Proceedings of the European Conference on Computer Vision, ECCV 2022*, vol. 13669, 2022, pp. 509–525.
- [50] Zhao, Xu, Zhang, Liu, Zhang, and Li, "DIDFuse: Deep image decomposition for infrared and visible image fusion," in *Proceedings of the IJCAI International Joint Conference on Artificial Intelligence*, vol. 2021-January, 2020, pp. 970–976.
- [51] Zhang, Zhang, Chen, and Wang, "RFNet: Multivariate long sequence time-series forecasting based on recurrent representation and feature enhancement," *Neural Networks*, vol. 181, p. 106800, 2025.
- [52] Qingyun, Dapeng, and Zhaokui, "Cross-Modality Fusion Transformer for Multispectral Object Detection," *arXiv e-prints*, p. arXiv:2111.00273, Oct. 2021.
- [53] Zhao, Bai, Zhang, Zhang, Xu, Lin, Timofte, and Van Gool, "CDDFuse: Correlation-driven dual-branch feature decomposition for multi-modality image fusion," in *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, 2023, pp. 5906–5916.
- [54] Yang, Liang, Li, and Zhang, "Multidimensional fusion network for multispectral object detection," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 35, no. 1, pp. 547–560, 2025.
- [55] Jocher and Qiu, "Ultralytics YOLO11," 2024. [Online]. Available: <https://github.com/ultralytics/ultralytics>
- [56] Zhao, Yuan, Jiang, Wang, and Wei, "Removal then selection: A coarse-to-fine fusion perspective for rgb-infrared object detection," *IEEE Transactions on Intelligent Transportation Systems*, pp. 1–16, 2025.